

# CoO-PILOT

Evidence-backed pre-clearance for Certificate of Origin claims. AI reads the documents; deterministic logic decides; a human signs off.  
SIH 2026 · IHSIH024 · TEAM NULL POINTERS · PROTOTYPE

## THE PROBLEM

A preferential-tariff claim under a trade agreement is checked by hand. An officer reads a commercial invoice, a packing list and a cost statement, checks they agree, and decides whether the goods meet the agreement's rules of origin. It is slow, inconsistent between officers, and leaves no structured record of *why* a claim was cleared. CoO-PILOT prepares the case and shows its working. It does not issue certificates.

## PIPELINE

|                                                    |                                                       |                                              |                                                   |                                         |                                     |                                       |
|----------------------------------------------------|-------------------------------------------------------|----------------------------------------------|---------------------------------------------------|-----------------------------------------|-------------------------------------|---------------------------------------|
| 1<br><b>Multimodal AI</b><br>Gemini · OCR + vision | 2<br><b>Structured evidence</b><br>field → source doc | 3<br><b>Reconciliation</b><br>cross-document | 4<br><b>Rules of origin</b><br>RVC + tariff shift | 5<br><b>Risk engine</b><br>triage score | 6<br><b>Autonomy gate</b><br>policy | 7<br><b>Officer</b><br>final decision |
|----------------------------------------------------|-------------------------------------------------------|----------------------------------------------|---------------------------------------------------|-----------------------------------------|-------------------------------------|---------------------------------------|

**Stages 1–2 are AI. Stages 3–6 are deterministic code. Stage 7 is a person.** The model reads; it never decides. A misread field surfaces as a reconciliation mismatch and routes to review rather than passing through unnoticed.

## ARCHITECTURE

| MODULE       | ROLE                                                                                                |
|--------------|-----------------------------------------------------------------------------------------------------|
| extraction/  | Gemini vision → structured JSON per document.<br>Retries on transient failure; caches on file hash. |
| rules/       | Reconciliation, origin criteria, risk scoring, decision policy.                                     |
| backend/     | FastAPI. Claims, documents, storage, orchestration, review workflow, audit.                         |
| console.html | Officer review console.                                                                             |

Each module is reached through a thin adapter, so any one can be swapped without touching the others. The backend imports the extractor and the rules engine by name and degrades honestly if either is absent.

## STACK

| LAYER  | CHOICE                                       |
|--------|----------------------------------------------|
| API    | FastAPI · Pydantic · Uvicorn                 |
| AI     | Google Gemini (multimodal)                   |
| Data   | Supabase Postgres · 7 tables, JSONB payloads |
| Files  | Supabase Storage · PDF/PNG/JPG ≤ 20 MB       |
| Deploy | Docker · Render blueprint                    |
| Tests  | 119, no credentials required                 |

**Falls back, never fakes**  
With no database configured the API runs in memory; with no API key, extraction reports the failure rather than inventing values. Every degraded state is visible in the response.

## WHAT THE SYSTEM DECIDES

| OUTCOME               | WHEN                                                                    | WORKED EXAMPLE                                                                                                                                            |
|-----------------------|-------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>APPROVED</b>       | Documents reconcile <i>and</i> the origin criterion is met              | Invoice and packing list agree at 500 units; RVC 42.86% against the AIFTA 35% threshold; all non-originating materials change tariff sub-heading. Risk 0. |
| <b>PENDING_REVIEW</b> | Anything unresolved — a discrepancy, or origin that cannot be evaluated | Invoice says 500 units, packing list says 480. Origin still passes, but the documents disagree, so it is held for a human. Risk 40.                       |
| <b>REJECTED</b>       | Origin was evaluated and failed                                         | Same documents, non-originating value 3500: RVC 16.67%, below the threshold.                                                                              |

A document discrepancy never auto-rejects — it routes to a person, which is what an officer would do.

### The refusal case

An invoice and a packing list *cannot* establish origin. Regional value content needs a cost statement: FOB value plus the value and classification of every non-originating material — the information CAROTAR 2020 requires an importer to hold in Form I. Without it the engine returns `INSUFFICIENT_DATA` and names the missing fields rather than estimating a percentage. Saying "I don't know" is a feature, not a gap.

RULES OF ORIGIN — AND THEIR LIMITS

Thresholds are stored as data with a citation and source URL attached to each, and the engine reports which criterion it applied so a reviewer can check the working.

| AGREEMENT               | VALUE<br>CONTENT | TARIFF<br>SHIFT | BASIS                                                                                                 |
|-------------------------|------------------|-----------------|-------------------------------------------------------------------------------------------------------|
| AIFTA (ASEAN–<br>India) | ≥ 35% of FOB     | CTSH · 6-digit  | AIFTA Rules of Origin, Rule 4. Given effect in India by Notification 189/2009-Cus (N.T.), 31.12.2009. |
| SAFTA (non-LDC)         | ≥ 40% of FOB     | CTH · 4-digit   | SAFTA Rules of Origin, twin criteria.                                                                 |
| SAFTA (LDC)             | ≥ 30% of FOB     | CTH · 4-digit   | Ten points below the non-LDC threshold.                                                               |

Regional value content is computed by the indirect (build-down) method:  $(\text{FOB} - \text{non-originating value}) \div \text{FOB} \times 100$ . Goods declared wholly obtained skip both tests.

Stated limits

- **General rule only.** Product-specific rules override it for many tariff lines and are not implemented.
- **Thresholds verified against secondary sources** (national trade portals), not gazette text. Verify before operational use.
- **Risk weights are ours**, a triage aid with no statutory basis. The interface says so on screen.
- **Three agreement variants** of India's many.

API

| ENDPOINT                                          | PURPOSE                   |
|---------------------------------------------------|---------------------------|
| POST /claims                                      | Open a case               |
| POST /claims/{id}/documents                       | Upload a document         |
| PUT /claims/{id}/origin-declaration               | Attach the cost statement |
| POST /claims/{id}/verify                          | Run the full pipeline     |
| GET /claims/{id}/result                           | Complete evidence record  |
| GET /claims/review                                | Review queue              |
| POST /claims/{id}/approve · reject · request-info | Officer decision          |
| GET /claims/{id}/audit                            | Audit trail               |

Plus per-document extract, reconcile, and dashboard counts — 23 routes, OpenAPI at /docs. The console consumes the same API a third party would.

HONEST STATUS

**Working end to end.** Live Gemini extraction from PDF and image; deterministic reconciliation; origin evaluation with citations; risk triage; review workflow writing to a real audit trail; 119 passing tests; container builds and runs.

AUDITABILITY

Every claim carries a full trail — `claim_created`, `document_uploaded`, `extraction_started` / `completed` / `failed`, `reconciliation_completed`, `processing_completed`, and the officer's `approved` / `rejected` / `information_requested`. Credentials are never written to it.

Each extracted field records the document it came from, so any value in a decision can be traced back to the page it was read from. Officer decisions are stored with reviewer and comments.

**Not yet.** Product-specific rules. Authentication — the API is unauthenticated and suits a controlled demo, not deployment. Persistence is configured but was exercised in memory. Extraction has been tested on a small set of document layouts.